Tank Battle

James Smith - Game Design Document
Assignment #2
Due: 26 October 2026 (Monday)

Multi-Review Agent Kit and Peer Review Summary

After submitting the original Game Design Document (GDD) to the review kit, the kit identified several significant design deficiencies. Although the following examples represent only a subset of the findings, they highlight several critical areas that required further clarification and refinement:

- **Game Completion Conditions:** Clearly defining the conditions under which the game ends.
- **Scoring System:** Establishing how scores are calculated, tracked, and maintained throughout the game.
- **Tie-Breaking Mechanisms:** Defining the rules for resolving ties and determining the final winner.

I addressed the identified issues throughout the document while making several executive decisions to keep the project within a manageable scope and budget. However, the extensive revisions prompted by the review process nearly doubled the original estimated development budget.

Additionally, the initial peer review did not fully address controls for mobile devices. I intend to design the game for cross-platform accessibility, ensuring it can be played across desktop computers, tablets, and mobile devices. As a result, mobile-specific control schemes remain an area for further consideration and refinement.

Executive Summary

Tank Battle is a 2D, scrollable, top-down online tank war where four color teams raid each other's depots for enemy-colored Energy Cells, and the team with the most Cells in its depot after five rounds wins.

Up to 20 players share one large scrolling world. When you join, pick a tank color (Blue/Water, Red/Fire, White/Ice, or Gray/Rock); everyone with your color is your team, and every other color is an enemy. No lobby or tutorial to wait in: the moment a round starts, you are in the action. Team-color Auxiliaries fight for their team, while black Wreckers belong to no team and hunt whoever carries the most Cells or whoever they encounter in open terrain. Wreckers can also attack depots, but only two can fire on a depot; the rest sit and wait or attack other defenders. If a depot is destroyed, the team has to go and gather more Cells.

Each team also owns an element: Blue tanks can only cross water, Red can only fire, White can only ice, and Gray can only rock; those same terrains hide tanks from Wreckers. Cells only score once you carry them out into the open and back to your depot, so greed and safety pull against each other all round. All players, Wreckers, and Auxiliaries can cross grass (Green) and ground (dirt) (Brown).

- **Core loop:** raid an enemy depot, carry its Cells back to your own depot to store them, survive enemies and Wreckers, refuel, and go again, across 5 rounds of at most 7 minutes each (35 minutes maximum).
- **Core pillar:** the raid-and-store loop. The elements, Wreckers, and elimination exist to put pressure on it.
- **Win:** the team with the most Cells in its depot at the end of round 5, or the last team standing.
- **Loss:** a destroyed player is out until the next round; a team whose human players are all destroyed in a round is removed from the game.
- **Platform and audience:** a desktop web game with keyboard controls, for players who like short team raids. Players can join or leave between rounds, so no one has to commit to all 35 minutes.

Game Mechanics

Each round, you move, aim, and fire at tanks, steal Cells from enemy depots, and refuel at stations, all while staying within your color's terrain.

What the player does

Action	What you see and do
Join	Pick Blue, Red, White, or Gray. Your tank appears in the live world, and the round is already running. A short hint overlay explains the controls the first time you join.
Move and aim	Steer (WASD) across the scrolling map; the view follows your tank. Open ground is passable for everyone, but each element is a wall to every color except its own.
Fire	Shoot enemy tanks, Auxiliaries, and Wreckers (Space Bar). Every target has 5 energy bars, and 3 hits drain one bar.
Raid	Drive into an enemy depot and pick up Cells in that depot's color. Your carried-Cell count shows on screen and rises as you steal, but the Cells do not score until you store them.
Store	Carry the stolen Cells back to your team's depot to store them. Only stored Cells count toward your team's total, and Cells you are still carrying can be lost if you are destroyed.
Defend	Guard your depot, where your team stores its Cells, because the other three teams are coming for your color's Cells.
Refuel	Drive onto a refueling station to refill all 5 energy bars. Only online players can refuel, and you must wait 30 seconds between refuels.
Hide	Slip into your own element to break Wrecker line of sight and wait out a hunt; shells and Wreckers cannot enter it, but firing from inside reveals you for 5 seconds.

One world per color

Team	Can cross	Cannot cross	Hiding terrain (blocks Wrecker sight)
Blue	Water	Fire, ice, rocks	Water
Red	Fire	Water, ice, rocks	Fire
White	Ice	Water, fire, rocks	Ice (snow piles)
Gray	Rocks	Water, fire, ice	Rocks (rock mounds)

Survival and elimination

A tank is destroyed after 15 hits without a refuel (5 bars at 3 hits each) and sits out until the next round, watching a spectator view of any remaining team. A team is removed from the game only if it had at least one human player at the start of the round and all its human players are destroyed by the end of the round; if only one team remains, the game ends, and that team wins.

The round loop

1. The round starts at 7:00 on the clock, and all online players spawn and act immediately.
2. Teams raid enemy depots and carry the Cells back to their own depot to store them. A new Wrecker appears every 15 seconds, so up to 28 can join the hunt by the end of a round, but the number alive at once is capped (6 in round 1, rising to 12 by round 4).
3. Players trade fire, refuel, hide, and push again as the map gets more dangerous.
4. The round ends when the timer runs out or only one team has human players left standing, and stored Cells are tallied. Teams with no humans left standing are removed; destroyed players on other teams re-enter next round.
5. After round 5, the team with the most stored Cells wins, so a match lasts at most 35 minutes.

Design Decisions

- Each depot starts with 20 Cells of its own color. A team's score is the number of Cells in its depot when round 5 ends, including its own and stolen Cells, and Cells carry over between rounds. If teams tie, the team that stole the most Cells wins.
- A tank carries up to 5 Cells and picks up 1 Cell per second at an enemy depot. A destroyed carrier drops its unstored Cells where it falls, and any player can pick them up.
- Each depot has 30 hits of HP. When it is destroyed, its Cells spill onto the ground for anyone to pick up, and the depot rebuilds empty 60 seconds later.
- Depots and refueling stations sit on open ground that every color can reach, never inside an element. Each depot sits at the edge of its team's element, with 2 nearby refueling stations, so retreating into cover is always possible.
- Each player must wait 30 seconds between refuels. Auxiliaries refuel at their own team's stations.
- A round counts as a scored round only when at least two teams have human players. Otherwise, it runs as practice: at most 2 Wreckers, no score, and no elimination.
- The game is balanced for 8 to 20 human players, with Auxiliaries filling empty slots.
- Players who join mid-round spawn at once with full energy, but a player destroyed this round cannot rejoin until the next round. If all 20 slots are full, new players queue and spectate.
- The number of Wreckers alive at once is the round cap (6, 8, 10, 12, and 12 across the five rounds) when 8 or more humans are online, and 0.6 per human, rounded up, when fewer are online. Wreckers reset at the start of each round.
- Wreckers hunt only a tank carrying at least 3 unstored Cells, so a one-Cell decoy does not draw them.
- Shells and Wreckers cannot enter an element, so a tank hiding in its own element is safe until it fires. A tank that fires from inside is revealed and can be hit for 5 seconds. Refueling is only possible on open ground.
- Auxiliaries use the same 5-bar, 3-hit durability as Wreckers and players. A team with no human players at the start of a round is out of play for that round, and its Auxiliaries are removed with it, as happens whenever a team is removed.
- Team elimination counts human players only, not Auxiliaries, and is checked only at the end of each round. A disconnected player does not count as destroyed.
- Energy bars are a tank's health and fuel; Energy Cells are the score. Wreckers are rogue black tanks that raid for no team.
- Players can only refuel while online. A saved session ID recognizes a disconnected player and returns at their last energy level.

AI Architecture

Tank Battle has two kinds of AI tanks: Wreckers punish whoever is winning, and Auxiliaries keep every team in the fight even when few players are online.

Wreckers (black, no team)

A Wrecker looks like a black tank with 5 energy bars. It cannot refuel; a destroyed Wrecker returns at a different spot on the map, and a fresh one arrives every 15 seconds. Its mission is to destroy everyone else, starting with whoever carries the most Cells.

- **It hunts the Cell leader.** Whoever carries the most unstored Cells (at least 3) is the target, so a big haul forces a choice between hiding and running to your depot. Ties go to the nearest tank, and the hunted player sees a warning marker. If no one qualifies, Wreckers head for the depot of the team with the most stored Cells.
- **It attacks anyone it sees,** so open ground is dangerous for everyone.
- **It cannot see through water, fire, ice (snow piles), or rock (rock mounds),** so a tank in its own element is invisible until it fires. Players hide while a Wrecker passes, then dash out.
- **It wears down.** Without refueling, a Wrecker dies after 15 hits, so focused fire can clear it, but another arrives within seconds.

- **It can attack depots.** Only two Wreckers can fire on a depot; the rest wait or attack other defenders, and a destroyed depot forces the team to gather more Cells.

Auxiliaries (team color, fight for their team)

An Auxiliary looks like a player tank in your team's color so that you can tell friend and foe at a glance.

- **It defends and fights.** It shoots visible enemies and holds ground near its own depot, so raiders meet resistance even without human teammates nearby.
- **It obeys its color's terrain rules.** A Blue Auxiliary crosses water but stops at fire, ice, and rock, letting enemies outmaneuver it.
- **It fills small teams.** When fewer than 20 people are online, Auxiliaries fill empty slots on every team with at least one human.
- **It can be shot.** Clearing Auxiliaries opens a depot for raiding.

Agents that build the game

Sixteen single-purpose agents, each run in Claude Code, design, develop, and test Tank Battle. Each does exactly one job.

Agent	Its one job
Lead Agent	Hands each task to the right agent and checks that their pieces fit together.
Game Logic Agent	Writes the rules for energy bars, 3-hit damage, carrying, storing, and counting stolen Cells, elimination, and round scoring.
World Map Agent	Builds the scrolling map with depots, refueling stations, and the water, fire, ice, and rock zones, and owns depot HP.
Game Control Agent	Turns player input into tank movement, aiming, and firing.
Wrecker AI Agent	Writes how black tanks roam, hunt the Cell leader, and lose sight of tanks hiding in terrain.
Auxiliaries AI Agent	Writes how team-color AI tanks defend their depot and attack visible enemies.
Graphics Agent	Draws the tanks in the four team colors plus black for Wreckers, and the Cells, depots, stations, and terrain tiles with clearly marked edges.
UI Agent	Designs the energy bars, Cell counter, round timer, team color display, first-join hint overlay, spectator view, depot and station markers, team standings board, and "you are being hunted" marker.
Multiplayer Sync Agent	Keeps all 20 players seeing the same world and keeps the 7-minute round timer on the server.
Playtest Agent	Runs scripted matches and reports bugs, such as a round that runs past 7 minutes.
Collision and Terrain Agent	Decides what blocks what: each element as a wall to the other three colors, shells hitting tanks, terrain cutting Wrecker line of sight, and a tank that fires from inside an element being revealed.
Session and Join Agent	Handles picking a color, the 20-player cap, returning destroyed players at the next round, refueling for online players only, queuing players when the world is full, recognizing returning players by a saved session ID, and owning the one-page player state table.
Audio Agent	Creates the sounds for firing, hits, destruction, refueling, and each new Wrecker arriving.
Deployment Agent	Sets up and keeps the persistent online game server running.
Effects Agent	Animates explosions, shell impacts, and the moving water, fire, snow, and rock terrain.
Balance Agent	Tunes the numbers that shape play, such as 3 hits per bar, the 15-second Wrecker spawn, and the 7-minute round.

Technical Strategy

Tank Battle is built using Claude Code by the sixteen single-purpose agents listed under AI Architecture, with a combined budget of about 3.8 million tokens, run in four waves so the project stays inside API rate limits.

Agent roles and token budget

Each agent is its own Claude Code session. Opus 5.5 handles coordination and multiplayer sync, Sonnet 5.5 handles gameplay code and AI, and Haiku 5.5 handles simpler work. Budget is tokens per task times the number of tasks, input plus output.

Agent	Model	Tokens/Task	Budget (tokens)	Cost
Lead Agent	Opus 5.5	20,000	240,000 (12 tasks)	$1.92
Game Logic Agent	Sonnet 5.5	25,000	300,000 (12 tasks)	$1.20
World Map Agent	Sonnet 5.5	22,500	180,000 (8 tasks)	$0.72
Game Control Agent	Haiku 5.5	17,500	140,000 (8 tasks)	$0.03
Wrecker AI Agent	Sonnet 5.5	26,000	260,000 (10 tasks)	$1.04
Auxiliaries AI Agent	Sonnet 5.5	22,000	220,000 (10 tasks)	$0.88
Graphics Agent	Haiku 5.5	20,000	200,000 (10 tasks)	$0.04
UI Agent	Haiku 5.5	20,000	160,000 (8 tasks)	$0.03
Multiplayer Sync Agent	Opus 5.5	28,000	280,000 (10 tasks)	$2.24
Playtest Agent	Sonnet 5.5	20,000	320,000 (16 tasks)	$1.28
Collision and Terrain Agent	Sonnet 5.5	22,000	220,000 (10 tasks)	$0.88
Session and Join Agent	Sonnet 5.5	20,000	160,000 (8 tasks)	$0.64
Audio Agent	Haiku 5.5	15,000	120,000 (8 tasks)	$0.02
Deployment Agent	Sonnet 5.5	20,000	160,000 (8 tasks)	$0.64
Effects Agent	Haiku 5.5	20,000	160,000 (8 tasks)	$0.03
Balance Agent	Sonnet 5.5	20,000	160,000 (8 tasks)	$0.64
Total			**3,280,000**	**$12.24**

A 15% reserve (about 492,000 tokens) covers rework after playtesting, bringing the plan to roughly 3.8 million tokens and about $14.08. Costs use Anthropic's October 2026 list prices per million input/output tokens (Opus 5.5 $4/$20, Sonnet 5.5 $2/$10, Haiku 5.5 $0.10/$0.50; see Sources), assuming 3 input tokens per output token and no caching. The budget covers agent work only; the developer's own integration and review time is extra.

Work runs in four waves:

(1) Lead, Game Logic, World Map, Game Control, and Collision and Terrain;
(2) Wrecker AI, Auxiliaries AI, Multiplayer Sync, Session and Join, and Deployment;
(3) Graphics, UI, Audio and Effects;
(4) Playtest and Balance, with fixes paid from the reserve.

After Wave 2, a playable slice (one map, two teams, networking, and one Wrecker) is tested before Waves 3 and 4 begin, so the netcode built by the Multiplayer Sync Agent is tested early. If the budget runs short, cuts come in this order, and a feature is cut only if nothing built later depends on it: Effects, Audio, Balance tuning, the join queue, then Auxiliary behavior beyond defending the depot. Real players playtest the slice after Wave 2 and the full game after Wave 4; the Graphics and Audio Agents create original sprites and sounds so no outside art is needed; and matchmaking and private rooms are out of scope. The plan runs 8 weeks: Wave 1 in weeks 1 and 2, Wave 2 in weeks 3 and 4 with the slice playtest at the end of week 4, Wave 3 in weeks 5 and 6, and Wave 4 in weeks 7 and 8, with about 10 hours of the developer's own integration time each week.

API constraints

- **Each call sees only its own brief.** An agent's context holds its spec, its own code, and the shared game-state sheet (position, team color, energy bars, Cells carried, terrain underfoot, plus depot state, round timer, and Wrecker spawn data), capped near 30,000 tokens. That sheet is why the Wrecker AI Agent can write "hunt the Cell leader".
- **There is no memory between calls.** Each session ends with a handoff note of 500 words or less to the Lead Agent.
- **Rate limits cap requests and tokens per minute.** Sixteen agents running at once would hit them, so we run no more than five together, which is why we split the work into waves.
- **The live game cannot call the API.** With up to 12 Wreckers alive at once, deciding 10 times per second, that's about 120 calls per second, each taking seconds, so Wrecker and Auxiliary behavior is fixed, rule-based code on the game server.
- **Server target.** The Multiplayer Sync Agent aims for about 20 state updates per second per player, and the 20-player cap keeps this affordable. For 20 players, about 60 entities per update at 20 updates per second comes to roughly 2.3 Mbit/s outbound, which a single small server can handle.

Tools

- Frontend: Angular, Geckos.io (client)
- Backend: TypeScript, Node.js/ExpressJS, Geckos.io (server)
- Game Framework: Phaser
- Netcode: the server is authoritative for movement, hits, and Cells; clients predict their own movement; Wreckers find paths with A* on the tile grid
- Hosting: a single small UDP-capable cloud server, which Geckos.io requires